

Research



Cite this article: Scauzillo RC, Ferkin MH. 2022 Do male meadow voles (*Microtus pennsylvanicus*) always prefer and signal to 'popular' females? The role of social information context. *Biol. Lett.* **18**: 20220017. <https://doi.org/10.1098/rsbl.2022.0017>

Received: 11 January 2022

Accepted: 22 February 2022

Subject Areas:
behaviour

Keywords:
context, scent marking, social information, rodent, preference

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Electronic supplementary material is available online at <https://doi.org/10.6084/m9.figshare.c.5872074>.

Animal behaviour

Do male meadow voles (*Microtus pennsylvanicus*) always prefer and signal to 'popular' females? The role of social information context

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For males, the behaviour of rival conspecifics can provide vital information about finding reproductively available females, and which have high-quality phenotypes. While valuable, this information represents sperm competition risk and even the potential for rejection by females. Thus, males should show flexibility in signalling behaviours towards females based on social information acquired with emphasis on the context of that information. We examined male meadow vole scent-marking behaviours and preferences for females based on previously obtained social information across different contexts. Social information context resulted in increased scent marking depending on the social odour pairing. Males scent marked more near a female whose odour had associated with three older rivals than one younger rival. But males marked equally toward a female whose odour had associated with three younger rivals and a female whose odour had associated with one older rival. This demonstrates that social odours are not of all equal value and that males can distinguish differences. Males then use these differences to tailor their signalling towards potentially high-quality phenotype females thereby maximizing their reproductive opportunities.

1. Introduction

Social information, and the audience effect, involving mate choice presents a crossroads for males. Males can gain valuable information about females and high-quality phenotypes, but it comes at the cost of sperm competition and potential rejection. Even with these costs, males have demonstrated the propensity to copy the mate choices of other males [1–3], though the number of studies remains limited. Furthermore, males are influenced by the presence of other males, altering their behaviours around and preferences for potential mates [4–6]. This change of behaviour is often considered 'deception' to draw a rival male away from a preferred female [7]. Overall, this indicates that males are aware of the costs of generating social information.

Social information will vary contextually due to phenotypic variation among individuals within a population. Social information context has influenced male preferences and behaviours [8,9]. Male guppies (*Poecilia reticulata*) will associate more with females accompanied by two drab coloured males than females accompanied by two colourful males [8]. An audience effect on male Atlantic mollies (*Poecilia mexicana*) was dependent on prior knowledge of the audience male. Focal males altered their preference when the audience male had been perceived as sexually active compared to sexually inactive [9].

Studies involving social information and mate choice have mainly been with fish and birds, but mammals are subject to social information through scent marking. Scent marking is a primary form of communication for

terrestrial mammals [10,11], conveys a wide range of information [12–14], and is considered to be cheat-proof [15]. Additionally, scent marks persist in the environment allowing conspecifics to investigate and interact with the mark(s) [12,16]. Interactions commonly elicited by these signals are over-marking and, sometimes, adjacent marking, [11,17] which then creates spatio-temporally varying contexts of social information. Therefore, an individual's own scent marking and preferences may be augmented differently based on the context of scent marks they encounter.

Meadow voles (*Microtus pennsylvanicus*) have a promiscuous mating system as females take up mutually exclusive territories while males wander large, shifting home ranges in search of reproductively available females [18,19]. Therefore, voles rely on scent marking to find and secure mating opportunities. Most scent marking, including over-marking, occurs in runway systems within their habitat used by both sexes [20]. Conspecifics can encounter and respond to other conspecifics' scent marks creating opportunities for an abundance of social information in these runways. When over-marking occurs between two males, females generally show a preference of the top scent donor [17]. When over-marking does not occur, females have been found to demonstrate preference based on age [21], hormone titres [22] and spatial ability and aggressiveness [23]. Additionally, females have adapted coercion prevention [24]. Males have also demonstrated preferences for female odours based on similar criteria [21,22]. Males also show a preference for the odour of 'popular' females, females whose odour associated with three or five, but not zero, one or two male odours, over that of a novel female [25]. This raises the question, does the context of rival male numbers and phenotypes matter?

This study was conducted to determine whether the context of previously encountered social information varied by rival age (older versus younger) and number of rivals (three versus one) affects scent-marking behaviour towards and preferences for females by males. We predict that males will scent mark more toward, and prefer, females who had associated with three older rivals over one younger rival or females who had associated with one older rival over three younger rivals.

2. Methods

Meadow voles used in this study were descendants of wild-caught voles housed at the University of Memphis under long photoperiod (14 L : 10 D, lights on at 07.00 h CST) which simulates the breeding season daylength. All voles were housed individually from the age of 35 days in cages (27 × 16.5 × 12.5 cm) that were lined with Sani-Chips bedding (P. J. Murphy Forest Products, Montville, NJ, USA) and provided water and food (Teklad Rodent Diet, No. 8640, Harlan, Madison, WI, USA) ad libitum.

The experiment was broken into two phases: a social odour information phase and a scent marking/preference phase. In the social odour phase, one group of 31 males received the soiled bedding (1 g per odour donor) of three older rivals with one female's bedding and the soiled bedding of one younger rival with a different female's bedding. Another group of 31 males received the soiled bedding of three younger rivals with one female's bedding and the soiled bedding of one older rival with another female's bedding. Focal males were between 150 and 400 days old. Females used were between

180 and 380 days of age corresponding to previous findings of male preferences [21]. We used mesh tea balls (2-inch diameter, X-Chef, Charlotte, NC) for odour delivery. Rivals' ages differed from the focal male by two and four months (younger or older). Soiled bedding was collected from odour donors on the day of the test; odour donors had been on the bedding for a minimum of 48 h. Two mesh tea balls were hung in the home cage of the focal male, based on group assignment, on the opposite side from where the male made his nest, spaced approximately 8 cm apart (electronic supplementary material, S1), and left for 30 min. After 30 min the mesh tea balls were removed, and the next phase commenced. Time between these phases was 2–3 min.

The second phase was a 10-min scent marking/preference test conducted in a modified T-shaped arena, arms and stem were 25 × 13 × 16 cm (L × W × H), the bottom was lined with white photocopy paper (electronic supplementary material, S2). At the end of each arm a small, clear acrylic tube connected the arena to a standard cage containing the females used as odour donors during the social odour phase; one female per side. Each acrylic tube contained a piece of wire mesh that prevented physical contact but allowed the transfer of olfactory and auditory signals. Focal males were placed in the stem of the arena for a 2-min acclimation period, a solid door prevented the male from exploring the arena. After 2 min the door was removed, and the male was allowed to explore the full arena. All tests were video recorded for preference analysis. After 10 min the male was returned to his home cage. The photocopy paper that lined the top of the arena was removed for scent marking analysis; this paper was allowed to dry for a minimum of 10 min before analysis. Scent marks were viewed in a dark room using a handheld UV illuminator (UVGL-58 Handheld UV Lamp, Upland, CA, USA); under UV light urine scent marks appear as blue streaks or spots. The photocopy paper from the stem section was not used for analysis and discarded. The arena, female cages and mesh tea balls were cleaned with 70% ethanol between tests. Females were used for multiple tests but no more than two times per group. Videos were analysed for the time spent by the male in each arm near the female.

We analysed the number of scent marks separately for each group using a generalized linear mixed models (GLMM) with a negative binomial distribution. In the model the fixed effect was the side of the arena and the random effect was the individual. We established a preference by dividing the time spent near the female whose odour associated with the three rivals by the total time spent associating with both females; this was analysed using a one sample *t* test. Significance was determined at $p < 0.05$. Statistical analyses were completed using R v. 4.1.0 [26] in RStudio v. 1.4.1717 [27]; we used the packages 'lme4' [28] and 'MASS' [29] to run our GLMMs.

3. Results

We found that male meadow voles scent marked towards a female whose odour associated with three older rivals (mean ± s.e.; 39.00 ± 3.93 scent marks) more than a female whose odour associated with one younger rival (30.32 ± 4.05 scent marks; $n = 31$, $z = 2.86$, $p < 0.01$; figure 1). There was a significant preference for the female whose odour associated with three older rivals than the female whose odour associated with one younger rival ($t_{30} = 2.37$, $p = 0.02$). Males deposited scent

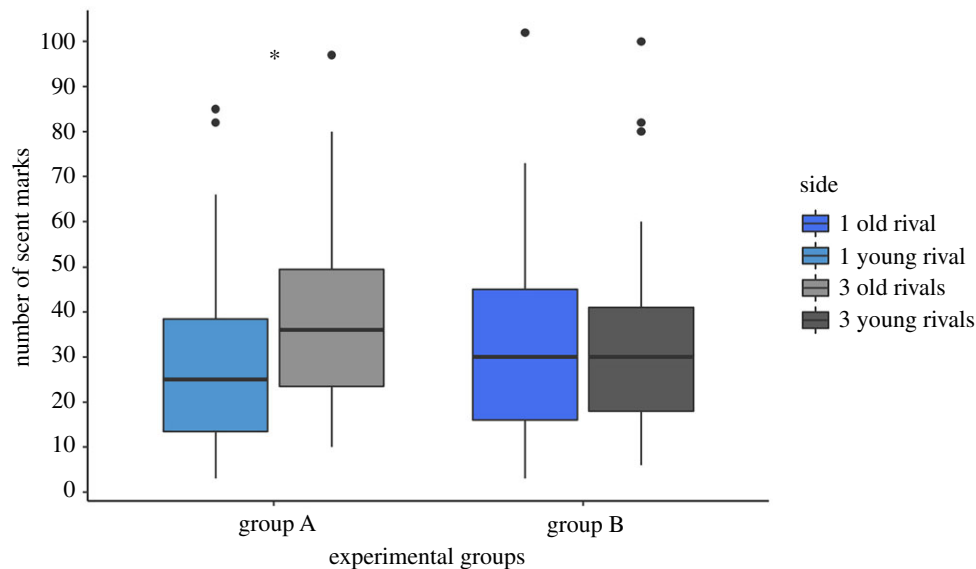


Figure 1. Boxplot (median, 25th and 75th percentiles, 95% confidence intervals, solid dots represent outliers) of the number of scent marks deposited by male meadow voles towards females. Group A represents the comparison between three old rivals and one young rival and group B represents the comparison between three young rivals and one old rival. Asterisk indicates significant difference $p < 0.01$.

marks in equal amounts toward a female whose odour had associated with three younger rivals (34.26 ± 4.02 scent marks) and a female whose odour had associated with one older rival (33.26 ± 4.13 scent marks; $n = 31$, $z = 0.54$, $p = 0.59$; figure 1). This was also reflected in the lack of preference by the males for either female ($t_{30} = 0.81$, $p = 0.43$).

4. Discussion

Male meadow voles' scent marking was influenced by the contextual differences of social information previously encountered. The flexibility seen may reflect the perceived value each social information instance provides and the inherent benefits and costs of acting on such information.

Males that signal toward females who have been solicited by multiple rivals compared to a single rival may suffer sperm competition and/or potential rejection. Female meadow voles prefer the odour of older males over younger males [21], so, a younger male signalling towards a particular female may be at a selective disadvantage. It is currently unknown exactly what changes occur to a male's odour as he ages or what females are preferring from older male odours. Male meadow voles will over-mark the scent marks of other males [30], and females show a preference for the top-scent donor of an over-mark [17,31,32]; this can be affected by characteristics of the signaller such as hormone titres [33]. With our experimental design, we did not present the opportunity for over-marking, but males may benefit from signalling toward a female through individual marks or by over-marking other male scent marks even if those males are older. Meadow voles have demonstrated multiple paternity [18,34], therefore, a male could still garner benefits siring even a small portion of a female's litter. Therefore, regardless of competition levels, males may see benefits from signalling toward females that have garnered multiple suitors.

The response by males when more rivals were younger compared to a single older rival presents an interesting notion about social information value. Males appear to value the information provided by three younger rivals equally to that of a single older rival as reflected with their

scent-marking behaviours. Males may view younger rivals as naive or less experienced, which could devalue their provided information compared to an older rival that may be seen as more experienced. As is the case with guppies and female mate choice copying [35]. Increasing the number of rivals may offset any notion of being naive, but still does not reach a threshold of being more valuable or reliable social information compared to information provided a single rival that may be considered experienced.

There is a lack of literature regarding prior social information encountered, context variation and its effects on signalling behaviours and mate choices by males. Most studies involving social information and male mate choice are from the perspective of an audience effect where the rival male(s) are physically present during the test [e.g. 4–6,9]. Few studies have examined male mate choice and/or signalling behaviours after acquiring social information where no rival(s) were present during the test [e.g. 2,3,8]. Clearly, males respond to social information, and do so differently across various social contexts. As olfactory communication is persistent, there may be many opportunities to acquire social information prior to making a choice. More research is needed to understand the use of previously encountered social information on male signalling behaviours and mate choices, especially for species that use olfactory communication.

Ethics. All animals were housed at the University of Memphis and used for this study followed the animal care protocol no. 0731, which was approved by the Institutional Animal Care and Use Committee at the University of Memphis.

Data accessibility. The dataset supporting this article, including a description file, has been uploaded as part of the electronic supplementary material [36].

Authors' contributions. R.C.S.: conceptualization, data curation, formal analysis, investigation, methodology, writing—original draft; M.H.F.: supervision, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Competing interests. We declare we have no competing interests.

Funding. The writing of this manuscript was supported by funds from the Department of Biological Sciences and the Jack H. Morris Distinguished Professorship to M.H.F.

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